

Problem

Find $F(s)$ if:

(a) $f(t) = 6e^{-t} \cosh 2t$

(b) $f(t) = 3te^{-2t} \sinh 4t$

(c) $f(t) = 8e^{-3t} \cosh tu(t-2)$

Step-by-step solution

Step 1 of 3

(a)

Consider the function.

$$f(t) = 6e^{-t} \cosh 2t$$

Recall Laplace transform pairs.

$$L[\cosh 2t] = \frac{s}{s^2 - 2^2}$$

$$L[e^{-t} \cosh 2t] = \frac{s+1}{(s+1)^2 - 4}$$

Use Laplace transform pairs to solve the function.

$$L[f(t)] = \frac{6(s+1)}{(s+1)^2 - 4}$$

$$F(s) = \frac{6(s+1)}{s^2 + 2s - 3}$$

Thus, Laplace transform of the function $f(t)$ is $\boxed{\frac{6(s+1)}{s^2 + 2s - 3}}$.

Step 2 of 3

(b)

Consider the function.

$$f(t) = 3te^{-2t} \sinh 4t$$

$$L[\sinh 4t] = \frac{4}{s^2 - 4^2}$$

$$L[t \sinh 4t] = -\frac{d}{ds} \left[\frac{4}{s^2 - 16} \right] = - \left(\frac{-4(2s)}{(s^2 - 16)^2} \right) = \frac{8s}{(s^2 - 16)^2}$$

$$L[e^{-2t} t \sinh 4t] = \frac{8(s+2)}{((s+2)^2 - 16)^2}$$

Use Laplace transform pairs to solve the function.

$$L(f(t)) = 3 \times \frac{8(s+2)}{((s+2)^2 - 16)^2}$$

$$F(s) = \frac{24(s+2)}{((s+2)^2 - 16)^2}$$

Thus, Laplace transform of the function $f(t)$ is $\boxed{\frac{24(s+2)}{((s+2)^2 - 16)^2}}$.

Step 3 of 3

(c)

Consider the function.

$$f(t) = 8e^{-3t} \cosh t \, u(t-2)$$

Hyperbolic function described as,

$$\cosh t = \frac{1}{2}(e^t + e^{-t})$$

Find the Laplace transform for $\cosh t \, u(t-2)$.

$$\begin{aligned} L[\cosh t \, u(t-2)] &= \frac{1}{2} \left[\int_2^\infty e^t e^{-st} dt + \int_2^\infty e^{-t} e^{-st} dt \right] \\ &= \frac{1}{2} \left[\int_2^\infty e^{-(s-1)t} dt + \int_2^\infty e^{-(s+1)t} dt \right] \\ &= \frac{1}{2} \left[\frac{-1}{s-1} e^{-(s-1)t} \Big|_2^\infty + \frac{-1}{s+1} e^{-(s+1)t} \Big|_2^\infty \right] \\ &= \frac{1}{2} \left[\frac{e^{-2(s-1)}}{s-1} + \frac{e^{-2(s+1)}}{s+1} \right] \end{aligned}$$

Find the Laplace transform for $e^{-3t} \cosh t \, u(t-2)$.

$$\begin{aligned} L[e^{-3t} \cosh t \, u(t-2)] &= \frac{1}{2} \left[\frac{e^{-2(s+3-1)}}{s+3-1} + \frac{e^{-2(s+3+1)}}{s+3+1} \right] \\ &= \frac{1}{2} \left[\frac{e^{-2(s+2)}}{s+2} + \frac{e^{-2(s+4)}}{s+4} \right] \end{aligned}$$

Use Laplace transform pairs to solve the function.

$$L(f(t)) = 8 \times \frac{1}{2} \left[\frac{e^{-2(s+2)}}{s+2} + \frac{e^{-2(s+4)}}{s+4} \right]$$

$$F(s) = \frac{4e^{-2(s+2)}}{s+2} + \frac{4e^{-2(s+4)}}{s+4}$$

Thus, Laplace transform of the function $f(t)$ is $\boxed{\frac{4e^{-2(s+2)}}{s+2} + \frac{4e^{-2(s+4)}}{s+4}}$.